STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 11: Thermodynamics

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. State the zeroth law.
2. Define internal energy.
3. First law with W done by system?
4. Meaning of adiabatic?
5. Meaning of isothermal?
6. Can a cyclic heat engine be 100% efficient?
7. Ideal gas equation?
8. Carnot efficiency?

Section B 2-mark questions - 6 x 2 = 12 marks

9. At constant volume, 500 J heat added: work?
10. Gas absorbs 800 J and does 300 J work: DeltaU?
11. Carnot T_h=600K,T_c=300K: efficiency?
12. Ideal-gas isothermal process: DeltaU?
13. mol with C_v=20 J/mol K and DeltaT=5 K: DeltaU?
14. Isothermal versus adiabatic?

Section C 3-mark questions - 4 x 3 = 12 marks

15. Gas absorbs 1000 J and does 400 J work: DeltaU?
16. Carnot T_h=500K,T_c=300K: eta?
17. Isobaric gas expands .01 to .03 m^3 at 100 kPa. Work?
18. Why no real cyclic engine converts all heat to work?

Section D 4-mark questions - 2 x 4 = 8 marks

19. Explain reversible and irreversible processes with examples.
20. Engine absorbs 1500 J from 600K, rejects to 300K; actual work=500J. Find Carnot limit, actual efficiency and Qout.